



Dhirubhai Ambani University
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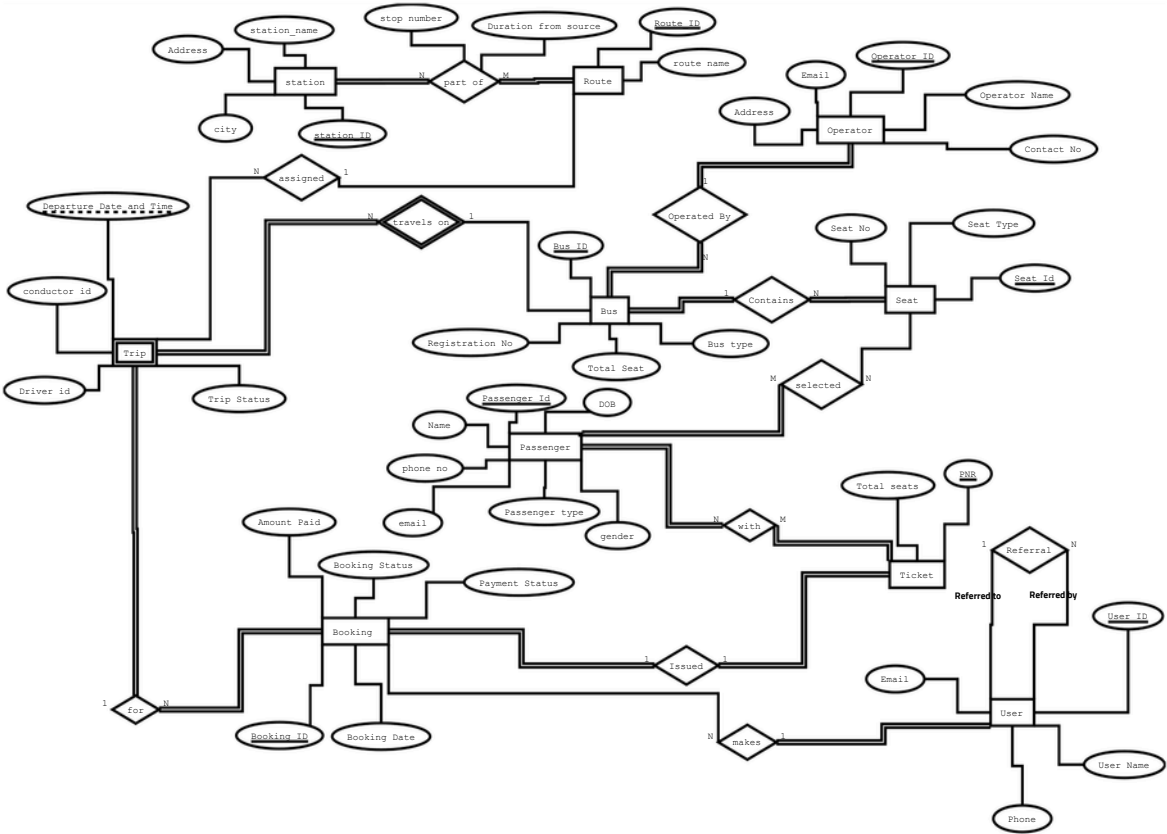
Database ERD & Relational Schema

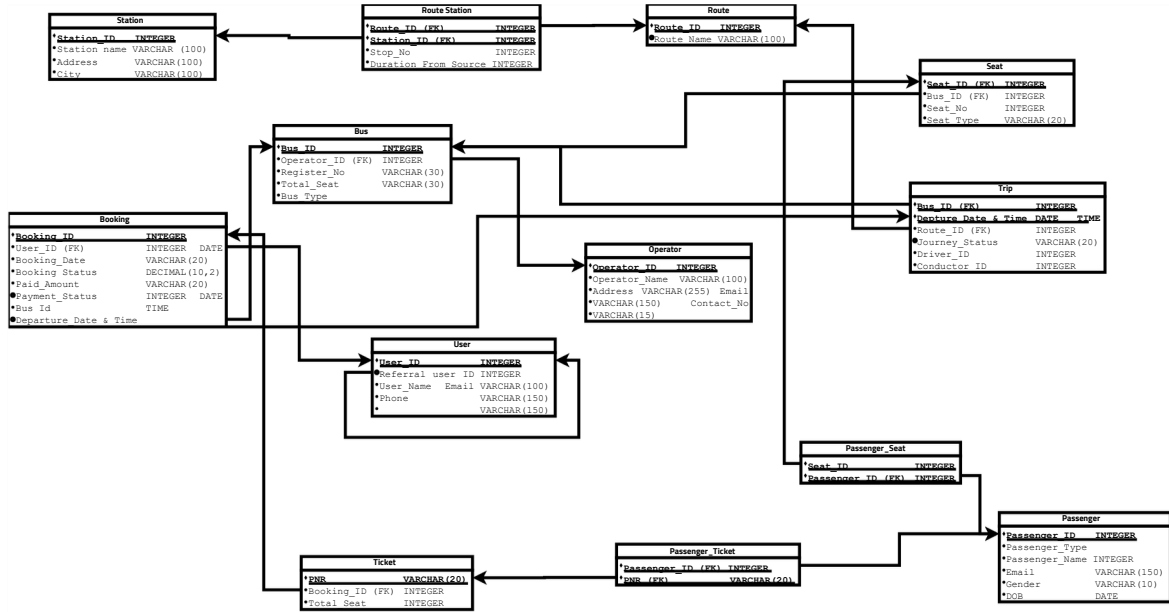
Course: IT214-Database management system
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Project Team

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ER Diagram





GoBus — Minimal FD Set

1) Bus

- BusID $\rightarrow$ RegNo
 - BusID $\rightarrow$ TotalSeats
 - BusID $\rightarrow$ BusType
 - BusID $\rightarrow$ OperatorID
 - RegNo $\rightarrow$ BusID
-

2) Operator

- OperatorID $\rightarrow$ Operator_Name
 - OperatorID $\rightarrow$ Address
 - OperatorID $\rightarrow$ Email
 - OperatorID $\rightarrow$ ContactNo
 - Email $\rightarrow$ OperatorID
 - ContactNo $\rightarrow$ OperatorID
-

- 3) **Route**
- RouteID $\rightarrow$ RouteName
 - RouteName $\rightarrow$ RouteID
-

4) Station

- StationID $\rightarrow$ StationName
 - StationID $\rightarrow$ Address
 - StationID $\rightarrow$ City
 - {StationName, City} $\rightarrow$ StationID
-

5) RouteStation

- {RouteID, StopNo} $\rightarrow$ StopNo
 - {RouteID, StationID} $\rightarrow$ DurationFromSource
 - {RouteID, StopNo} $\rightarrow$ StationID
-

6) Seat

- SeatID $\rightarrow$ BusID
- SeatNo $\rightarrow$ SeatNo
- SeatID $\rightarrow$ SeatType
- {BusID, SeatNo} $\rightarrow$ SeatID

7) User • UserID →
ReferralID • UserID →
Username • UserID →
Email • UserID → Phone
• Email → UserID •
Phone → UserID

8) Passenger

- PassengerID → PassengerType
 - PassengerID → PassengerName
 - PassengerID → Email
 - PassengerID → Gender
 - PassengerID → DOB
-

9) Ticket

- PNR → BookingID
 - PNR → TotalSeat
 - BookingID → PNR
-

10) Booking

- BookingID → UserID
 - BookingID → BookingDate
 - BookingID → BookingStatus
 - BookingID → PaidAmount
 - BookingID → PaymentStatus
 - BookingID → BusID
 - BookingID → DepartureDateAndTime
-

11) Trip

- {BusID, DepartureDateAndTime} → RouteID
- {BusID, DepartureDateAndTime} → JourneyStatus
- {BusID, DepartureDateAndTime} → DriverID
- {BusID, DepartureDateAndTime} → ConductorID

BCNF Check — GoBus

Rule: For every functional dependency $X \rightarrow Y$, X must be a Superkey of the relation.

1) Bus

Attributes: BusID, RegNo, TotalSeats, BusType, OperatorID

Finding Candidate Keys:

- $\text{BusID} \rightarrow \text{RegNo, TotalSeats, BusType, OperatorID}$ → BusID determines all attributes → **BusID is a Candidate Key**
- $\text{RegNo} \rightarrow \text{BusID} \rightarrow \text{BusID} \rightarrow \text{RegNo, TotalSeats, BusType, OperatorID} \rightarrow \text{RegNo}$ determines all attributes → **RegNo is a Candidate Key**

Candidate Keys: {BusID}, {RegNo}

Checking each FD for BCNF:

- $\text{BusID} \rightarrow \text{RegNo} \rightarrow \text{BusID}$ is a Candidate Key
- $\text{BusID} \rightarrow \text{TotalSeats} \rightarrow \text{BusID}$ is a Candidate Key
- $\text{BusID} \rightarrow \text{BusType} \rightarrow \text{BusID}$ is a Candidate Key
- $\text{BusID} \rightarrow \text{OperatorID} \rightarrow \text{BusID}$ is a Candidate Key
- $\text{RegNo} \rightarrow \text{BusID} \rightarrow \text{RegNo}$ is a Candidate Key

Bus is in BCNF

2) Operator

Attributes: OperatorID, Operator_Name, Address, Email, ContactNo

Finding Candidate Keys:

- $\text{OperatorID} \rightarrow \text{Operator_Name, Address, Email, ContactNo} \rightarrow \text{OperatorID}$ determines all attributes → **OperatorID is a Candidate Key**
- $\text{Email} \rightarrow \text{OperatorID} \rightarrow \text{OperatorID} \rightarrow \text{all attributes} \rightarrow \text{Email}$ determines all attributes → **Email is a Candidate Key**
- $\text{ContactNo} \rightarrow \text{OperatorID} \rightarrow \text{OperatorID} \rightarrow \text{all attributes} \rightarrow \text{ContactNo}$ determines all attributes → **ContactNo is a Candidate Key**

Candidate Keys: {OperatorID}, {Email}, {ContactNo}

Checking each FD for BCNF:

- $\text{OperatorID} \rightarrow \text{Operator_Name} \rightarrow \text{OperatorID}$ is a Candidate Key
- $\text{OperatorID} \rightarrow \text{Address} \rightarrow \text{OperatorID}$ is a Candidate Key
- $\text{OperatorID} \rightarrow \text{Email} \rightarrow \text{OperatorID}$ is a Candidate Key
- $\text{OperatorID} \rightarrow \text{ContactNo} \rightarrow \text{OperatorID}$ is a Candidate Key
- $\text{Email} \rightarrow \text{OperatorID} \rightarrow \text{Email}$ is a Candidate Key
- $\text{ContactNo} \rightarrow \text{OperatorID} \rightarrow \text{ContactNo}$ is a Candidate Key

Operator is in BCNF

3) Route

Attributes: RouteID, RouteName

Finding Candidate Keys:

- $\text{RouteID} \rightarrow \text{RouteName} \rightarrow \text{RouteID}$ determines all attributes $\rightarrow$ **RouteID is a Candidate Key**
- $\text{RouteName} \rightarrow \text{RouteID} \rightarrow \text{RouteName}$ determines all attributes $\rightarrow$ **RouteName is a Candidate Key**

Candidate Keys: {RouteID}, {RouteName}

Checking each FD for BCNF:

- $\text{RouteID} \rightarrow \text{RouteName} \rightarrow \text{RouteID}$ is a Candidate Key
- $\text{RouteName} \rightarrow \text{RouteID} \rightarrow \text{RouteName}$ is a Candidate Key

Route is in BCNF

4) Station

Attributes: StationID, StationName, Address, City

Finding Candidate Keys:

- $\text{StationID} \rightarrow \text{StationName, Address, City} \rightarrow \text{StationID}$ determines all attributes $\rightarrow$ **StationID is a Candidate Key**
- $\{\text{StationName, City}\} \rightarrow \text{StationID} \rightarrow \text{StationID} \rightarrow \text{StationName, Address, City} \rightarrow \{\text{StationName, City}\}$ determines all attributes $\rightarrow$ **{StationName, City} is a Candidate Key**

Candidate Keys: {StationID}, {StationName, City}

Checking each FD for BCNF:

- $\text{StationID} \rightarrow \text{StationName} \rightarrow \text{StationID}$ is a Candidate Key
- $\text{StationID} \rightarrow \text{Address} \rightarrow \text{StationID}$ is a Candidate Key
- $\text{StationID} \rightarrow \text{City} \rightarrow \text{StationID}$ is a Candidate Key
- $\{\text{StationName}, \text{City}\} \rightarrow \text{StationID} \rightarrow \{\text{StationName}, \text{City}\}$ is a Candidate Key

Station is in BCNF

5) RouteStation

Attributes: RouteID, StopNo, StationID, DurationFromSource

Finding Candidate Keys:

- $\{\text{RouteID}, \text{StopNo}\} \rightarrow \text{StationID}, \text{DurationFromSource} \rightarrow \{\text{RouteID}, \text{StopNo}\}$ determines all attributes $\rightarrow \{\text{RouteID}, \text{StopNo}\}$ is a Candidate Key
- $\{\text{RouteID}, \text{StationID}\} \rightarrow \text{StopNo}, \text{DurationFromSource} \rightarrow \{\text{RouteID}, \text{StationID}\}$ determines all attributes $\rightarrow \{\text{RouteID}, \text{StationID}\}$ is a Candidate Key (Reasoning: A specific station on a specific route is assigned exactly one stop number)

Candidate Keys: $\{\text{RouteID}, \text{StopNo}\}$, $\{\text{RouteID}, \text{StationID}\}$

Checking each FD for BCNF:

- $\{\text{RouteID}, \text{StopNo}\} \rightarrow \text{StationID} \rightarrow \{\text{RouteID}, \text{StopNo}\}$ is a Candidate Key
- $\{\text{RouteID}, \text{StopNo}\} \rightarrow \text{DurationFromSource} \rightarrow \{\text{RouteID}, \text{StopNo}\}$ is a Candidate Key
- $\{\text{RouteID}, \text{StationID}\} \rightarrow \text{StopNo} \rightarrow \{\text{RouteID}, \text{StationID}\}$ is a Candidate Key
- $\{\text{RouteID}, \text{StationID}\} \rightarrow \text{DurationFromSource} \rightarrow \{\text{RouteID}, \text{StationID}\}$ is a Candidate Key

RouteStation is in BCNF

6) Seat

Attributes: SeatID, BusID, SeatNo, SeatType

Finding Candidate Keys:

- $\{\text{BusID}, \text{SeatNo}\} \rightarrow \text{SeatID}, \text{SeatType} \rightarrow \{\text{BusID}, \text{SeatNo}\}$ determines all attributes $\rightarrow \{\text{BusID}, \text{SeatNo}\}$ is a Candidate Key
- $\text{SeatID} \rightarrow \text{BusID}, \text{SeatNo}, \text{SeatType} \rightarrow \text{SeatID}$ determines all attributes $\rightarrow \text{SeatID}$ is a Candidate Key (Reasoning: A primary key ID perfectly identifies every detail of the row, including the seat number)

Candidate Keys: {SeatID}, {BusID, SeatNo}

Checking each FD for BCNF:

- $\{BusID, SeatNo\} \rightarrow SeatID \rightarrow \{BusID, SeatNo\}$ is a Candidate Key
- $\{BusID, SeatNo\} \rightarrow SeatType \rightarrow \{BusID, SeatNo\}$ is a Candidate Key
- $SeatID \rightarrow BusID \rightarrow SeatID$ is a Candidate Key
- $SeatID \rightarrow SeatNo \rightarrow SeatID$ is a Candidate Key
- $SeatID \rightarrow SeatType \rightarrow SeatID$ is a Candidate Key

Seat is in BCNF

7) User

Attributes: UserID, ReferralID, Username, Email, Phone

Finding Candidate Keys:

- $UserID \rightarrow ReferralID, Username, Email, Phone \rightarrow UserID$ determines all attributes $\rightarrow$
UserID is a Candidate Key
- $Email \rightarrow UserID \rightarrow UserID \rightarrow$ all attributes $\rightarrow$ Email determines all attributes $\rightarrow$
Email is a Candidate Key
- $Phone \rightarrow UserID \rightarrow UserID \rightarrow$ all attributes $\rightarrow$ Phone determines all attributes $\rightarrow$
Phone is a Candidate Key

Candidate Keys: {UserID}, {Email}, {Phone}

Checking each FD for BCNF:

- $UserID \rightarrow ReferralID \rightarrow UserID$ is a Candidate Key •
- $UserID \rightarrow Username \rightarrow UserID$ is a Candidate Key •
- $UserID \rightarrow Email \rightarrow UserID$ is a Candidate Key •
- $UserID \rightarrow Phone \rightarrow UserID$ is a Candidate Key •
- $Email \rightarrow UserID \rightarrow Email$ is a Candidate Key •
- $Phone \rightarrow UserID \rightarrow Phone$ is a Candidate Key

User is in BCNF

8) Passenger

Attributes: PassengerID, PassengerType, PassengerName, Email, Gender, DOB

Finding Candidate Keys:

- PassengerID → PassengerType, PassengerName, Email, Gender, DOB → PassengerID determines all attributes → **PassengerID is a Candidate Key**

Candidate Keys: {PassengerID}

Checking each FD for BCNF:

- PassengerID → PassengerType → PassengerID is a Candidate Key
- PassengerID → PassengerName → PassengerID is a Candidate Key
- PassengerID → Email → PassengerID is a Candidate Key
- PassengerID → Gender → PassengerID is a Candidate Key
- PassengerID → DOB → PassengerID is a Candidate Key

Passenger is in BCNF

9) Ticket

Attributes: PNR, BookingID, TotalSeat

Finding Candidate Keys:

- PNR → BookingID, TotalSeat → PNR determines all attributes → **PNR is a Candidate Key**
- BookingID → PNR → PNR → BookingID, TotalSeat → BookingID determines all attributes → **BookingID is a Candidate Key**

Candidate Keys: {PNR}, {BookingID}

Checking each FD for BCNF:

- PNR → BookingID → PNR is a Candidate Key
- PNR → TotalSeat → PNR is a Candidate Key
- BookingID → PNR → BookingID is a Candidate Key

Ticket is in BCNF

10) Booking

Attributes: BookingID, UserID, BookingDate, BookingStatus, PaidAmount, PaymentStatus, BusID, DepartureDateAndTime

Finding Candidate Keys:

- BookingID → UserID, BookingDate, BookingStatus, PaidAmount, PaymentStatus, BusID, DepartureDateAndTime → BookingID determines all attributes → **BookingID** is a **CandidateKey**

Candidate Keys: {BookingID}

Checking each FD for BCNF:

- BookingID → UserID → BookingID is a Candidate Key
- BookingID → BookingDate → BookingID is a Candidate Key
- BookingID → BookingStatus → BookingID is a Candidate Key
- BookingID → PaidAmount → BookingID is a Candidate Key
- BookingID → PaymentStatus → BookingID is a Candidate Key
- BookingID → BusID → BookingID is a Candidate Key
- BookingID → DepartureDateAndTime → BookingID is a Candidate Key

Booking is in BCNF

11) Trip

Attributes: BusID, DepartureDateAndTime, RouteID, JourneyStatus, DriverID, ConductorID

Finding Candidate Keys:

- {BusID, DepartureDateAndTime} → RouteID, JourneyStatus, DriverID, ConductorID
→ {BusID, DepartureDateAndTime} determines all attributes → **{BusID, DepartureDateAndTime}** is a **CandidateKey**

Candidate Keys: {BusID, DepartureDateAndTime}

Checking each FD for BCNF:

- {BusID, DepartureDateAndTime} → RouteID → {BusID, DepartureDateAndTime} is a Candidate Key
- {BusID, DepartureDateAndTime} → JourneyStatus → {BusID, DepartureDateAndTime} is a Candidate Key
- {BusID, DepartureDateAndTime} → DriverID → {BusID, DepartureDateAndTime} is a Candidate Key
- {BusID, DepartureDateAndTime} → ConductorID → {BusID, DepartureDateAndTime} is a Candidate Key

Trip is in BCNF